

THE AI TOOLBOX · COLLEGE GUIDE

Choose Your Major — Then Aim Your Research

What do universities like **Princeton** actually offer? And how does one research project turn an application from "interested" into **proven**?

Merit Point Academy · 选择你的专业 · 45 min · press **S** notes · → next ·  to comment

WHY THIS MATTERS NOW

A research topic is **evidence**, not decoration

Selective universities don't just ask "are you smart?" They ask "**what will you study here, and why should we believe you?**" Almost every top application has a "**why this major**" essay or an intended field of study.

Say it

"I want to study neuroscience." — anyone can write this sentence.

Show it

"I ran a study measuring reaction-time and attention." — now the claim has a spine.

Aim it

Pick the **major first**, and your research topic can point straight at it — one story, not two.

The loop of this hour: choose a major direction → let it **aim** your course research topic → the finished project becomes the **strongest evidence** that you belong in that major. This is why we're spending a class on majors inside a research course.

Major, concentration, minor — what's the difference?

Major = Concentration 主修专业

Your main field — most of your advanced courses and your senior thesis live here. Princeton calls it a **concentration**. You usually **declare it at the end of sophomore year**, but your application already signals the direction.

Minor / Certificate 辅修 / 证书

A smaller, focused add-on — often **interdisciplinary** and open to *any* major. Example: Princeton's **Statistics & Machine Learning (SML)** minor lets a Politics or Biology major add real data skills.

Two Princeton degrees: the **A.B.** (Bachelor of Arts — humanities, social sciences, most sciences) and the **B.S.E.** (Bachelor of Science in Engineering). **Computer Science is offered as both** — same field, two flavors. In total Princeton has **37 concentrations** and **50+ minors & certificates**.

All 37 concentrations, in four divisions

Humanities

- Architecture
- Art & Archaeology
- Classics
- Comparative Literature
- East Asian Studies
- English
- French & Italian
- German
- History
- Music
- Near Eastern Studies
- Philosophy
- Religion
- Slavic Langs & Lits
- Spanish & Portuguese

Social Sciences

- African American Studies
- Anthropology
- Economics
- Linguistics
- Politics
- Psychology
- Public & International Affairs
- Sociology

Natural Sciences

- Astrophysical Sciences
- Chemistry
- Computer Science (A.B.)
- Ecology & Evolutionary Biology
- Geosciences
- Mathematics
- Molecular Biology
- Neuroscience
- Physics

Engineering · B.S.E.

- Chemical & Biological Eng.
- Civil & Environmental Eng.
- Computer Science (B.S.E.)
- Electrical & Computer Eng.
- Mechanical & Aerospace Eng.
- Operations Research & Financial Eng.

Source: Princeton Admission — Degrees & Departments. Computer Science appears in both columns (A.B. and B.S.E.), so 38 listings = 37 distinct concentrations. A few fields straddle two divisions.

Majors your AI research can point at

The data & ML skills you're building in this course support **many** majors — not just Computer Science. For each, a sample research topic *you could actually do this term*:

Computer Science

Studies: algorithms, ML, systems. **Topic ▶** "Can a small CNN read handwritten digits as well as a big one?"

Oper. Research & Fin. Eng. (ORFE)

Studies: optimization, data, risk. **Topic ▶** "Which features best predict bike-share demand at rush hour?"

Neuroscience

Studies: brain + computation. **Topic ▶** "Do reaction-time patterns predict attention lapses?"

Molecular Biology

Studies: genes, cells, data. **Topic ▶** "Can gene-expression clusters separate two tissue types?"

Economics

Studies: markets, causal data. **Topic ▶** "How does study time relate to test scores across schools?"

Psychology

Studies: behavior + statistics. **Topic ▶** "Does background music change reading-comprehension scores?"

And the umbrella: whatever you major in, the **Statistics & Machine Learning minor** formally certifies the exact skills this course teaches. Your project can be the "substantial application of machine learning" that minor requires.

Quick check · tap an answer

You love medicine and you're building a machine-learning project in this course. Which plan makes the strongest college application?

Major in Computer Science, because that's where AI lives, and drop the medicine interest

Major in Molecular Biology or Neuroscience, and aim your ML project at a health question — one story

Pick whatever major is easiest to get into, research topic doesn't matter

Don't declare any interest; keep all options open so nothing ties you down

THE MECHANISM · IN ONE PICTURE

Interest → Major → Research topic

1

Interest

What you can't stop reading about

2

Major

The field that studies it, at a real university

3

Research topic

A question inside that field you can answer
this term

4

Application

Essay + project that all point one way

Worked example A

Loves the brain → **Neuroscience** → "Does sleep predict memory-test scores?" → project + essay agree.

Worked example B

Loves markets → **Economics / ORFE** → "What predicts second-hand phone prices?" → quantitative, on-theme.

Worked example C

Loves language → **Comparative Lit / Linguistics** → "Can a model tell two authors apart?" → NLP meets the humanities.

TRY IT · MAJOR EXPLORER

Pick an interest — meet its majors

Click a row. Each opens to Princeton majors in that direction and a research angle you could take **using this course's tools**.

▶  Biology, health & medicine

▶  Building software, robots & devices

▶  Money, decisions & data

▶  The mind & people

▶  Language, culture & history

▶  The physical world

The same interest, many names

Princeton is our worked example, but every university slices majors differently. Learn to **compare** — the label changes, the field doesn't:

MIT

Names majors by number — **Course 6** is Electrical Eng. & Computer Science, and **6-9** is Computation & Cognition (CS + brain).

Stanford

Has CS with an AI track, plus **Symbolic Systems** — a blend of CS, linguistics, psychology & philosophy.

Carnegie Mellon

A whole **School of Computer Science**, and the first US undergraduate **B.S. in Artificial Intelligence**.

Homework habit: for any dream school, find its "majors / fields of study" page and locate the 2–3 that fit your interest. The names differ; your job is to match the **content**, not the label.

THE HARD PART · TALK TO YOUR PARENTS

The parent conversation · 和父母的对话

This decision is a **family** one. Come prepared, not defensive. A 15-minute agenda:

Bring to the table

- **1–2 majors** you're drawn to, and *why* — in one sentence each
- Your **research topic** and how it points at that major
- What that major can **lead to** (careers, graduate paths)

Ask your parents

- What do you worry about with this path? 你担心什么？
- What matters most — stability, passion, income, impact?
- Can we pick a major that honors **both** our hopes?

If you disagree: a **minor** or a **double interest** is often the peace treaty — e.g. major in Economics (parents' comfort) + **SML minor** and an AI research topic (your passion). Coherent *and* reassuring.

Quick check · tap an answer

At Princeton, which statement is TRUE?

Every major leads to the same B.A. degree

Computer Science is offered as both an A.B. and a B.S.E.

You must declare your major before you apply, and can never change it

Machine learning can only be studied by Computer Science majors

Your "Major + Topic" one-pager

1. **Have the parent conversation** using the agenda two slides back. Note one thing you agreed on and one worry they raised.
2. **Pick 1–2 target majors** at a dream school (Princeton or any). Write the exact name and the division it's in.
3. **Aim your research topic:** in 3–4 sentences, explain how your course project would be **evidence** for that major.
4. **Bonus:** find that school's "fields of study" page and screenshot the 2–3 majors that fit you.

Submit & grading

Turn in the one-pager in the **Assignments** portal. Graded on **coherence**: does the major, the topic, and the "why" tell *one* story?



[Open Assignments · 作业](#)

Stuck choosing? Re-open the **Major Explorer** and start from the interest you couldn't stop reading about. Post a draft on any slide with  — your instructor will see it.

WRAP-UP

Three things to keep

- A university is a **menu of majors** — Princeton alone has **37 concentrations**; your job is to find the 1–2 that fit *you*
- **AI is a method, not one major** — carry it into biology, economics, psychology, language, physics; certify it with a minor like SML
- Pick the major, **aim the research**, and your application tells **one** convincing story — said *and* shown

Exit ticket: name the major you'd aim at, in one line. Post it in .

Next up · back to the toolbox — Class 05: Data Analysis Tools, Python, Gently.